

Garvit Sonawala

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RESEARCH INTERESTS

Theoretical and computational cosmology, with a focus on early-Universe physics, dark matter phenomenology, high-energy theory, and gravitational-wave cosmology. Interests include developing and analysing theoretical models of cosmic and particle phenomena, leveraging computational and machine-learning techniques for simulation, inference, and data analysis, while keeping the primary emphasis on advancing underlying physical theory.

EDUCATION

IISER Thiruvananthapuram - BS-MS (Integrated) in Physics 2022 – 2027
CGPA: 7.8 / 10

IIT Madras - BS in Data Science 2023 – Present
CGPA: 8.21 / 10

Delhi Public School, Varanasi - CBSE Class XII 2021
92.4%

AWARDS & RECOGNITION

GATE 2025 - Physics (PH) 2025
Qualified the Graduate Aptitude Test in Engineering (Physics) at national level.

RESEARCH EXPERIENCE

Minor Project in Chemistry IISER Thiruvananthapuram Jan 2026 – Present
Supervisor: Dr. R. S. Swathi
Developed and benchmarked Particle Swarm Optimization (PSO) on standard test functions, and applied it to determine minimum-energy configurations of $(B_{12})_n$ clusters ($N = 2-10$) using an improved Lennard-Jones and antiaromatic repulsion potential under a rigid-body approximation.

Student Intern Tata Institute of Fundamental Research (TIFR), Mumbai May 2025 – July 2025
Supervisor: Dr. Achamveedu Gopakumar
Developed gravitational-wave detection models for circular SMBHBs using 12-year PTA timing residuals (0PN–2PN). Implemented CNN, RNN, and Bayesian neural networks for waveform classification and chirp-mass estimation at low SNR.

Student Intern IIT Guwahati Jul – Sep 2024
Supervisor: Dr. Subhaditya Bhattacharya
Analysed the evolution of dark matter number density using the Boltzmann equation, deriving the yield evolution equation and studying freeze-out behaviour in the radiation-dominated early universe.

PRESENTATIONS

Exploring Dark Matter Freeze-Out ANVESHA 2024, IISER Thiruvananthapuram
Poster presentation on dark matter's cosmological significance, freeze-out behaviour, and observational evidence, based on the IITG internship work.

TECHNICAL SKILLS

Programming: Python, Java, SQL
Libraries / Tools: NumPy, Pandas, Matplotlib, Jupyter, PyTorch, scikit-learn, Git, $\LaTeX$
Machine Learning: CNN, RNN, Bayesian Neural Networks; supervised & unsupervised learning, regression, clustering
Database / Web: MySQL, PostgreSQL, Flask, SQLAlchemy

RELEVANT COURSEWORK

Physics: Classical Mechanics, Quantum Mechanics I & II, Electrodynamics, Statistical Mechanics, Special & General Relativity, Nuclear Physics, Astrophysics, Quantum Field Theory, Particle Physics
Mathematics: Linear Algebra, Multivariable Calculus, Probability & Statistics
Data Science: Machine Learning, Data Analysis & Visualization

VOLUNTEERING & LEADERSHIP

Core Committee Member, PSIT Physics Society, IISER TVM Aug 2024 – Present
Manage sponsorships, event logistics, merchandise, and content creation for departmental and outreach activities.